Lecture 8A: Visual Imitation

Jitendra Malik

Towards an Embodied AI, inspired by child development

Perception and Interaction

Language

456 A. M. TURING :

Instead of trying to produce a programme to simulate the adult mind, why not rather try to produce one which simulates the child's ? If this were then subjected to an appropriate course of education one would obtain the adult brain. Presumably the child-brain is something like a note-book as one buys it from the stationers. Rather little mechanism, and lots of blank sheets. (Mechanism and writing are from our point of view almost synonymous.) Our hope is that there is so little mechanism in the child-brain that something like it can be easily programmed. The amount of work in the education we can assume, as a first approximation, to be much the same as for the human child.

Turing (1950)
Computing Machinery
And Intelligence

The Development of Embodied Cognition: Six Lessons from Babies

Linda Smith & Michael Gasser

- Be multi-modal
- Be incremental
- Be physical
- Explore
- Be social
- Use language

BAIR
BERKELEY ARTIFICIAL INTELLIGENCE RESEARCH

SFV: Reinforcement Learning of Skills from Videos

Xue Bin Peng, Angjoo Kanazawa, Jitendra Malik, Pieter Abbeel, Sergey Levine

SIGGRAPH Asia 2018


Xue Bin Peng


Angjoo Kanazawa


Jitendra Malik


Pieter Abbeel

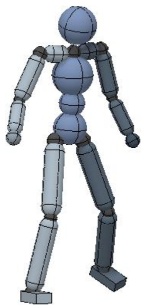

Sergey Levine

Instead… use video

YouTube

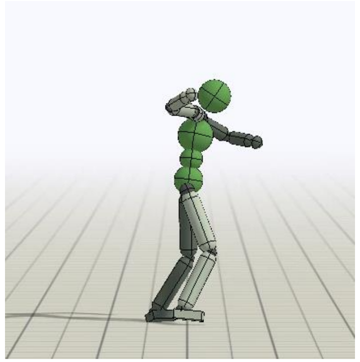
Deep RL based motion imitation

DeepMimic [Peng et al. SIGGRAPH 2018]



Character
In simulation

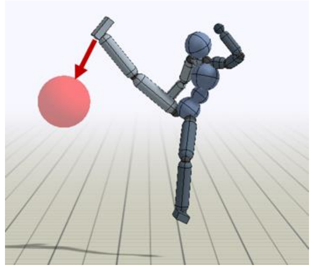
Need
Actions



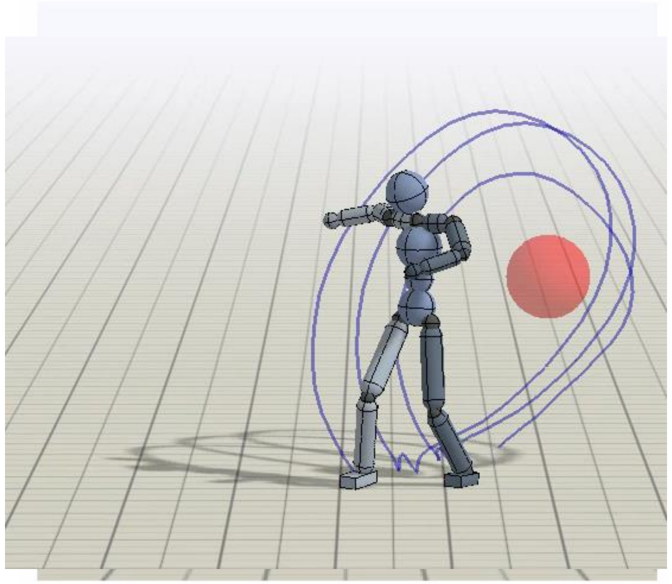
Reference Motion
(**Mocap**)

Only Recorded
joint angles

Reinforcement
Learning



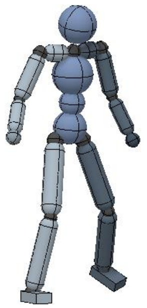
Task: Hit Target



Still requires reference motion from mocap

Deep RL based motion imitation

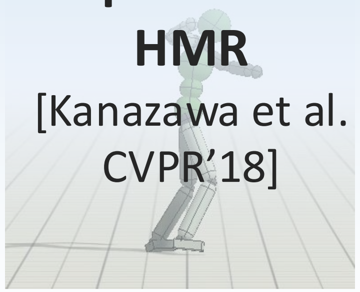
DeepMimic [Peng et al. SIGGRAPH 2018]



Character
In simulation

Need
Actions

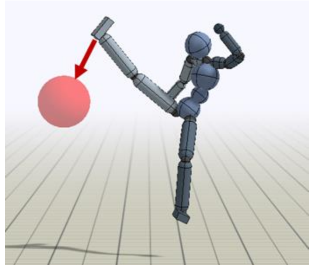
Replace with HMR
[Kanazawa et al. CVPR'18]



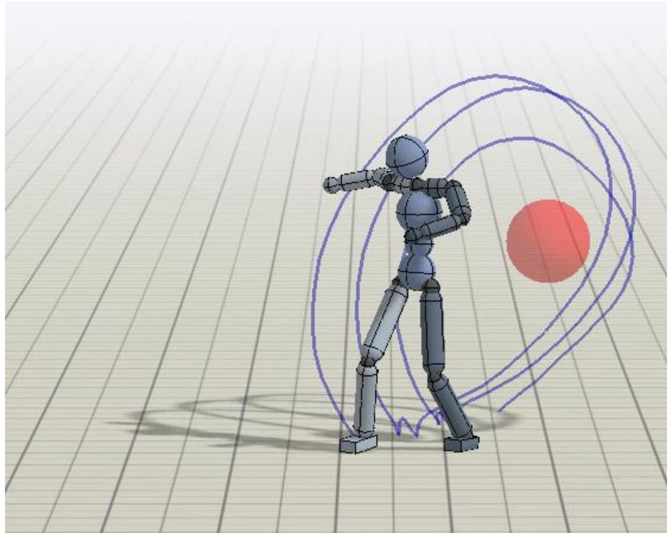
Reference Motion
(**Mocap**)

Only Recorded
joint angles

Reinforcement
Learning



Task: Hit Target



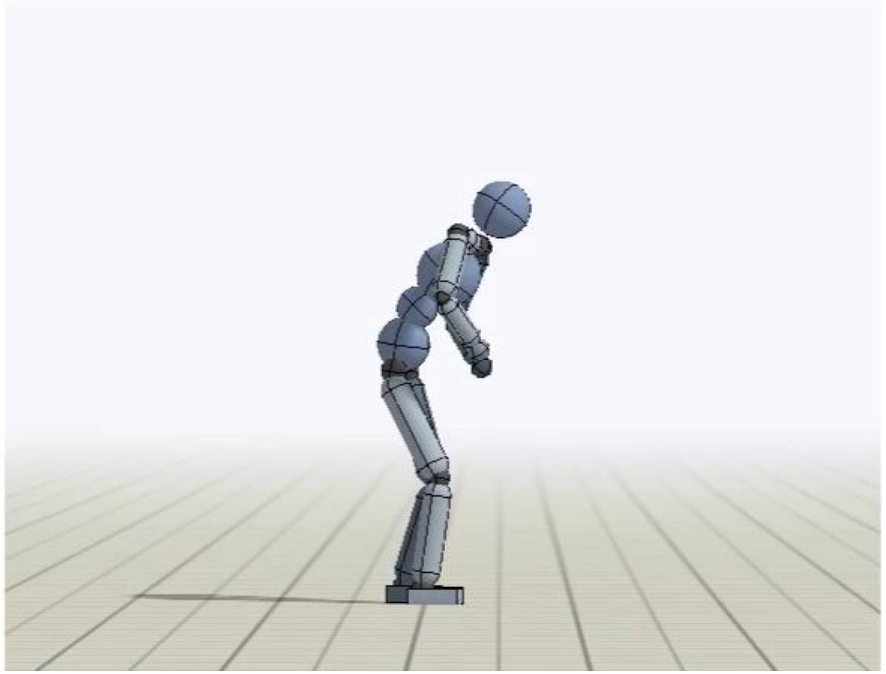
Learning Dynamic Skills from Videos

Video

HMR + temporal smoothing

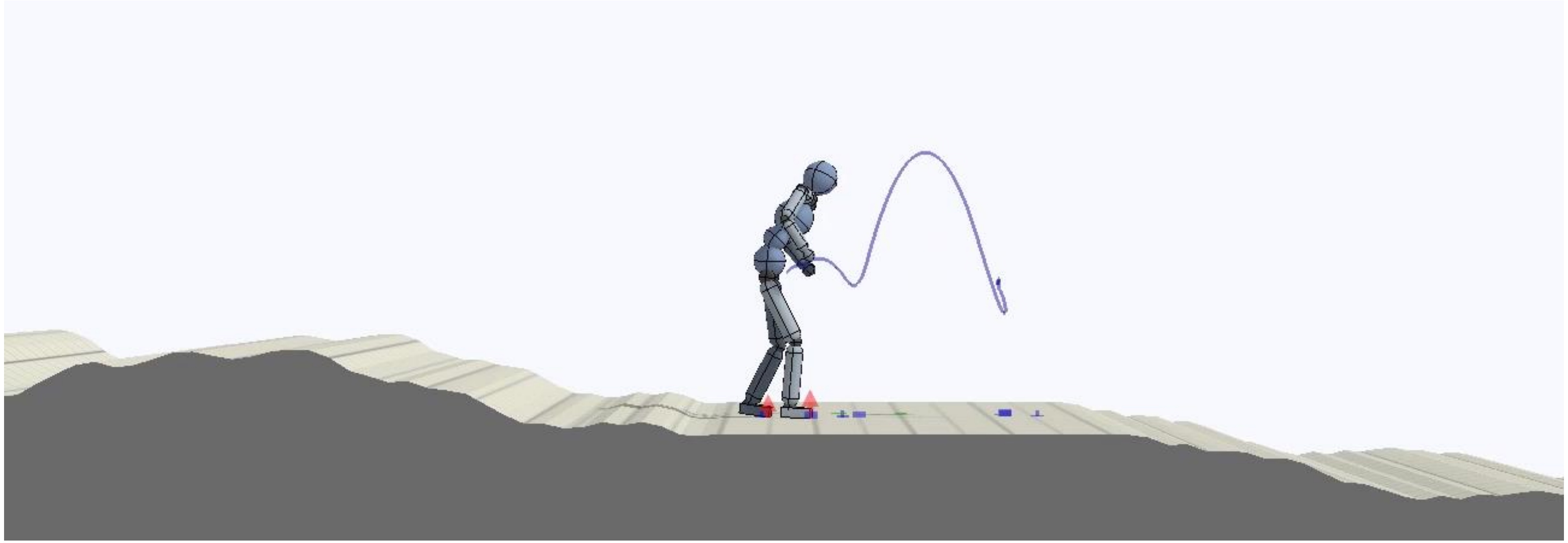
Use recovered 3D pose to train a physically simulated agent

Video



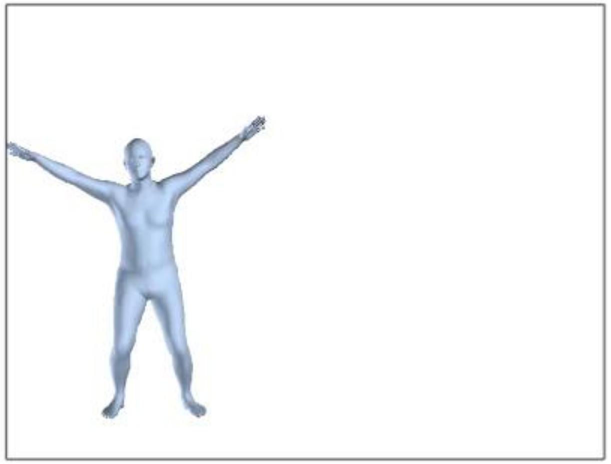
Simulation

Environment Retargeting



Humanoid: Cartwheel

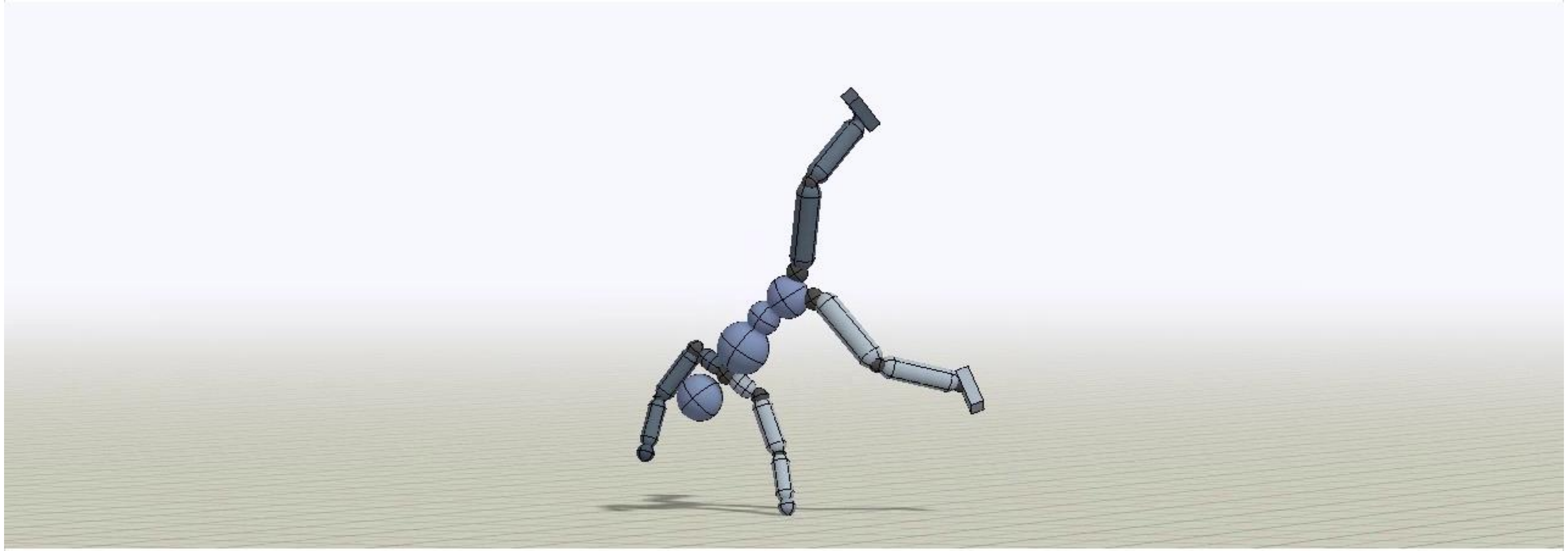
Video: Cartwheel B



Reference Motion

Policy

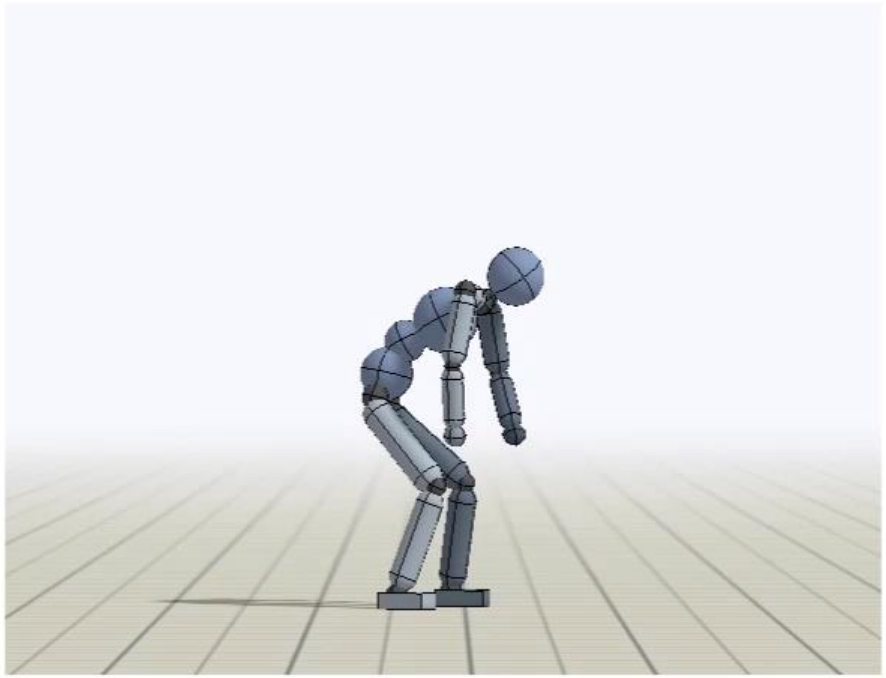
Robustness



The policies learn to recover from significant perturbations.

Humanoid: Frontflip

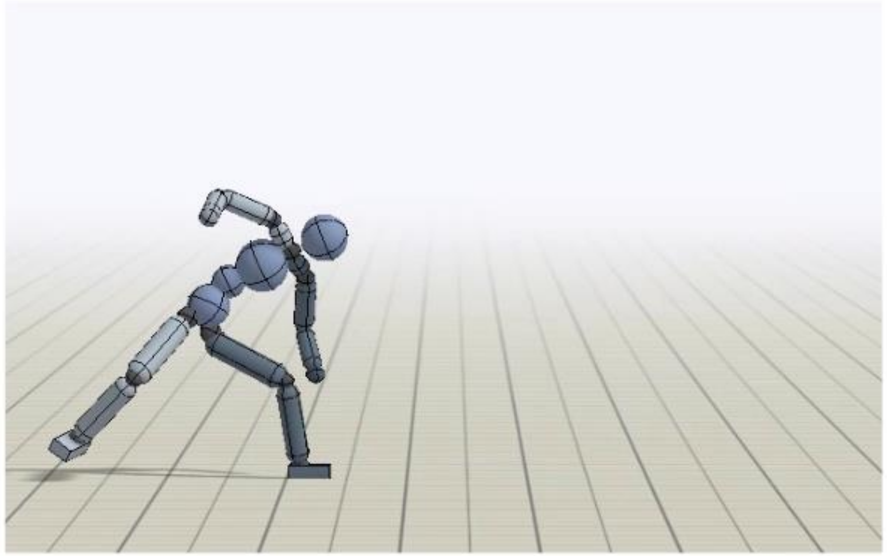
Video: Frontflip



Policy

Humanoid: Spin

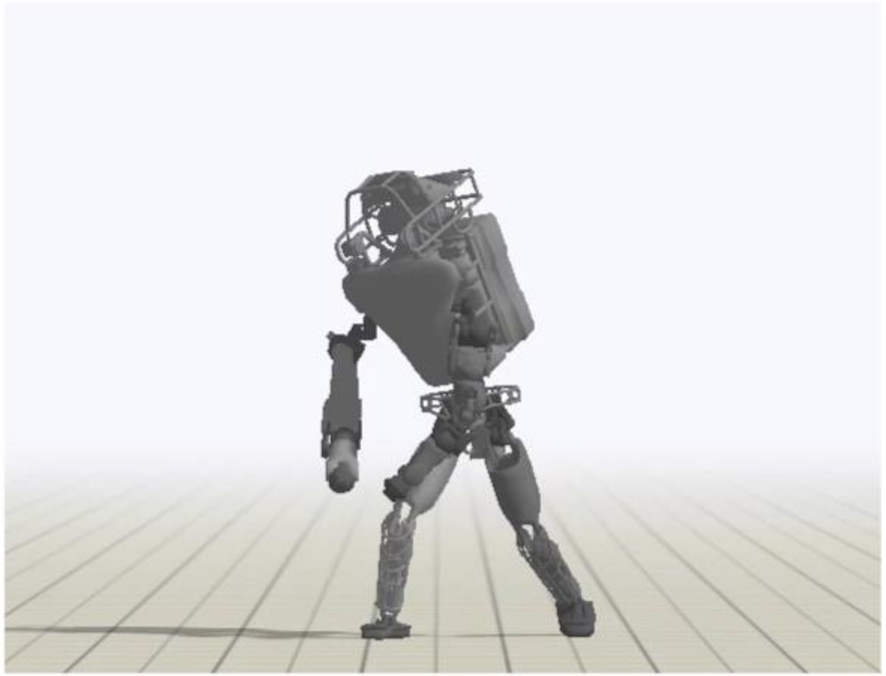
Video: Spin



Policy

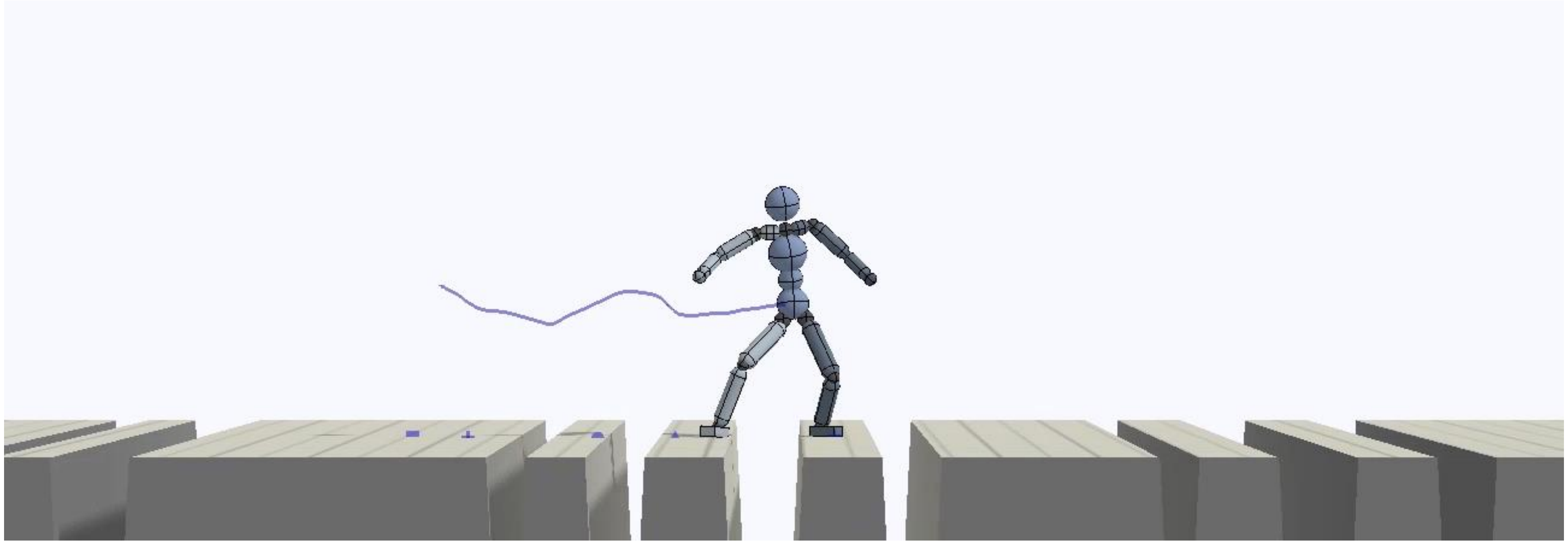
Atlas: Dance

Video: Dance



Policy

Environment Retargeting

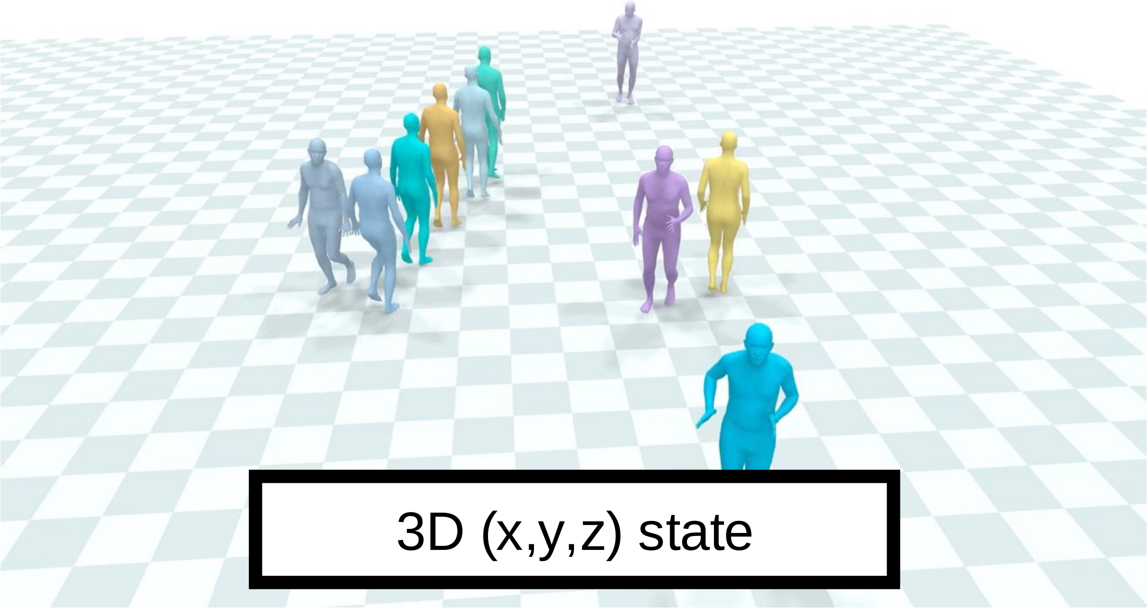


Reconstructing 4D Human Actions

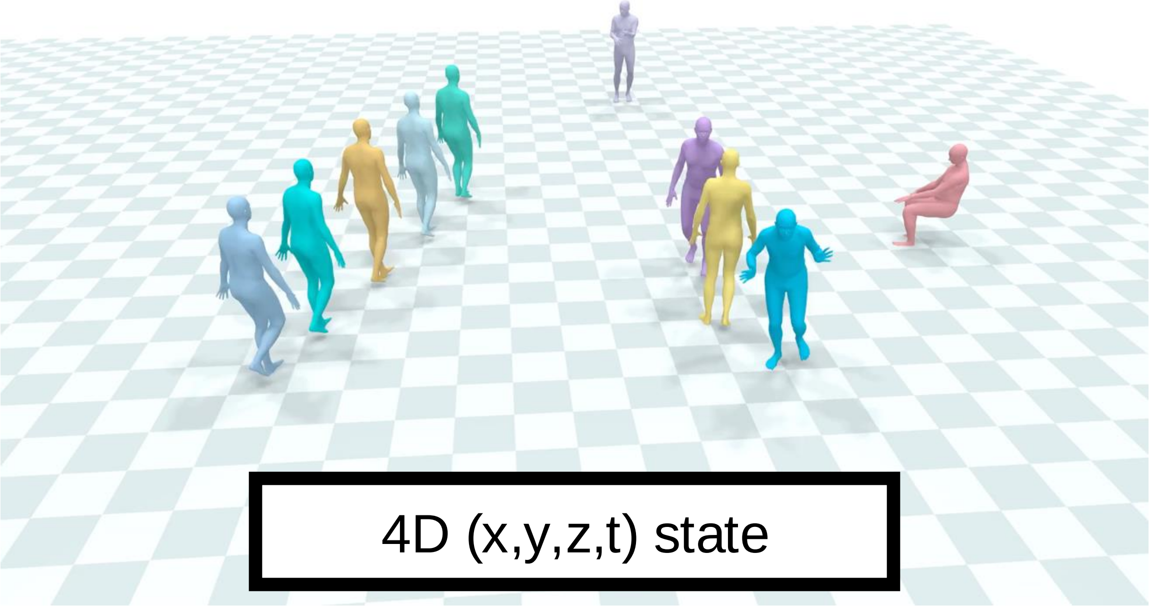
Jitendra Malik
UC Berkeley
FAIR, Meta

BERKELEY ARTIFICIAL INTELLIGENCE RESEARCH


2D (x,y) observation


3D (x,y,z) state


3D (x,y,t) observation


4D (x,y,z,t) state

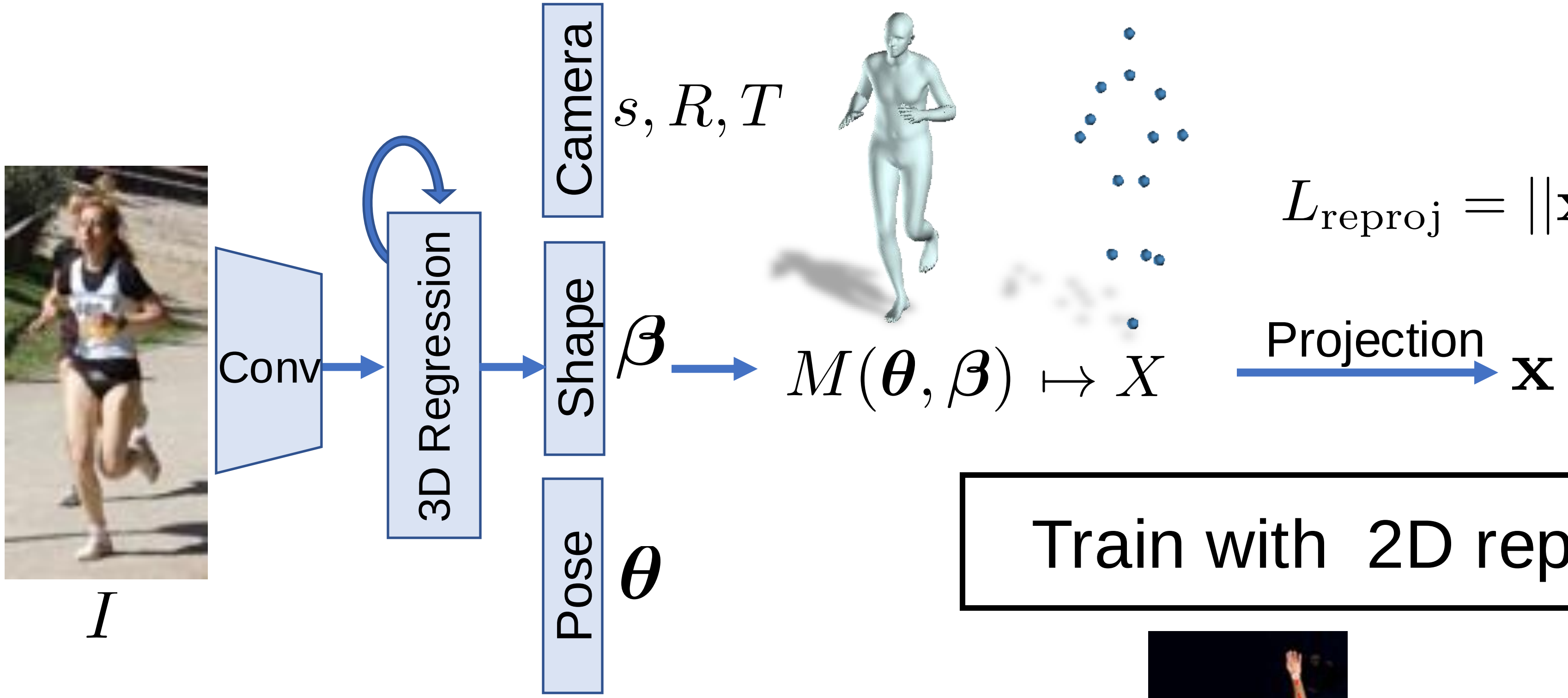
Humans in 4D: Reconstructing and Tracking Humans with Transformers

ICCV 2023

Shubham Goel Georgios Pavlakos Jathushan Rajasegaran
Angjoo Kanazawa Jitendra Malik

UC Berkeley

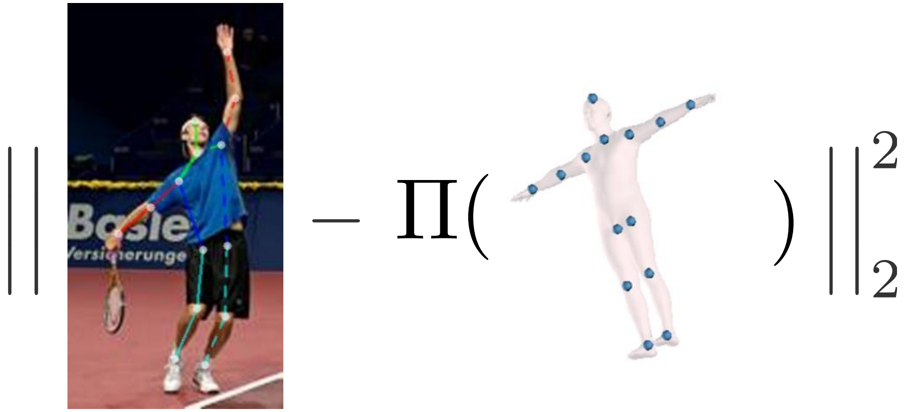
Human Mesh Recovery (HMR) 1.0



$\mathbf{x}^*$: 2D ground truth joints

$$L_{\text{reproj}} = ||\mathbf{x}^* - \mathbf{x}||$$

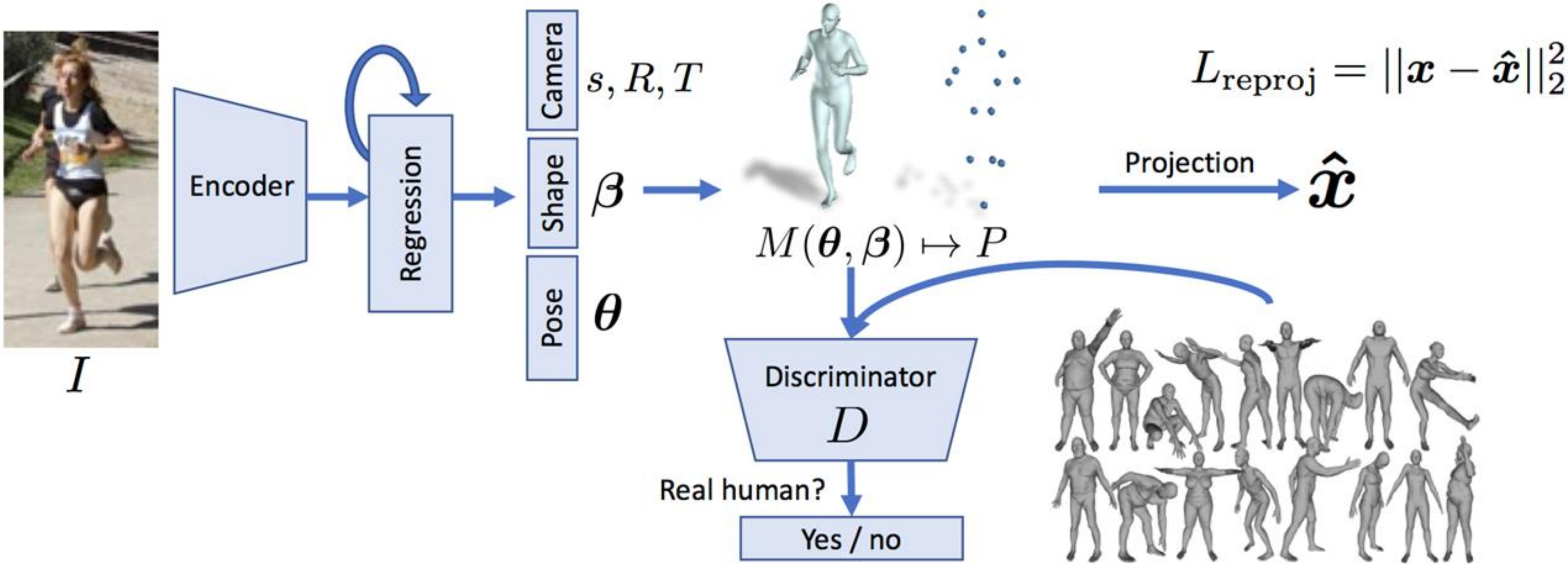
Train with 2D reprojection loss



Kanazawa, Black, Jacobs and Malik, CVPR 2018

Without any 2D-to-3D supervision…

Approach (2018)



Approach (2023)

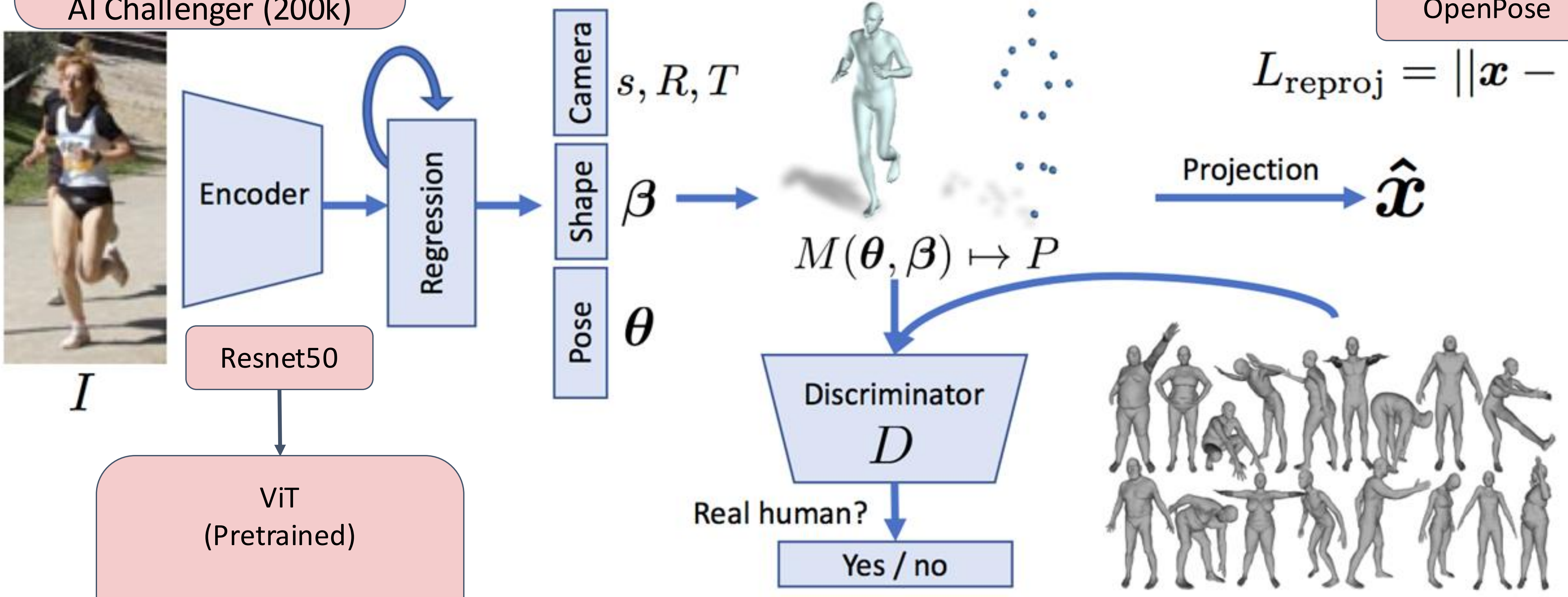
New Data:
AVA (60hr) +
InstaVariety (200hr) +
AI Challenger (200k)

Pseudo-GT 3D Fitting
(ProHMR)

ViTPose

OpenPose

$$L_{\text{reproj}} = ||\boldsymbol{x} - \hat{\boldsymbol{x}}||_2^2$$

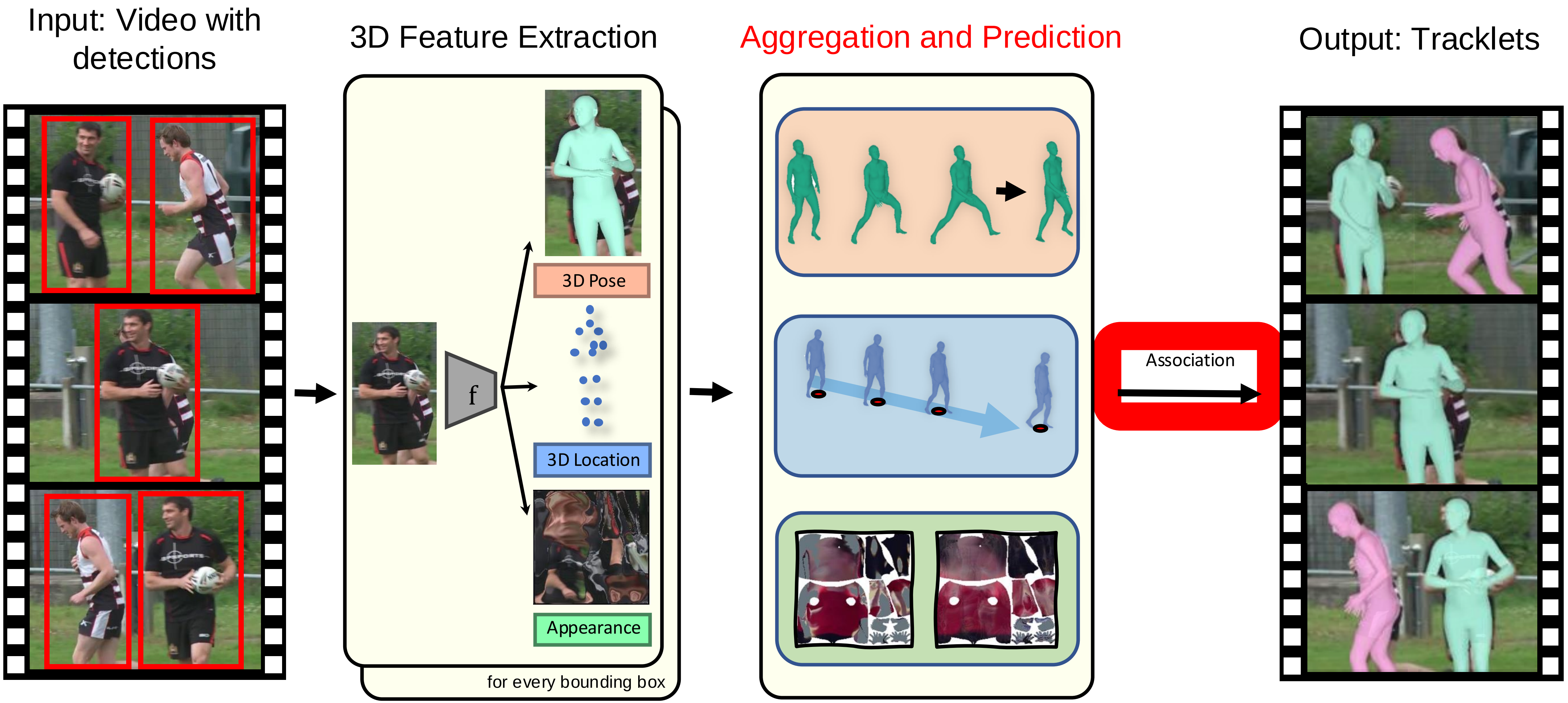


Resnet50

ViT
(Pretrained)

MAE ImageNet
2D-KeyPt COCO+...

PHALP: Tracking Humans by Predicting 3D Appearance, 3D Location & 3D Pose (CVPR 2022)



Rajasegaran, Pavlakos, Kanazawa, Malik
CVPR 2022

Insight #1

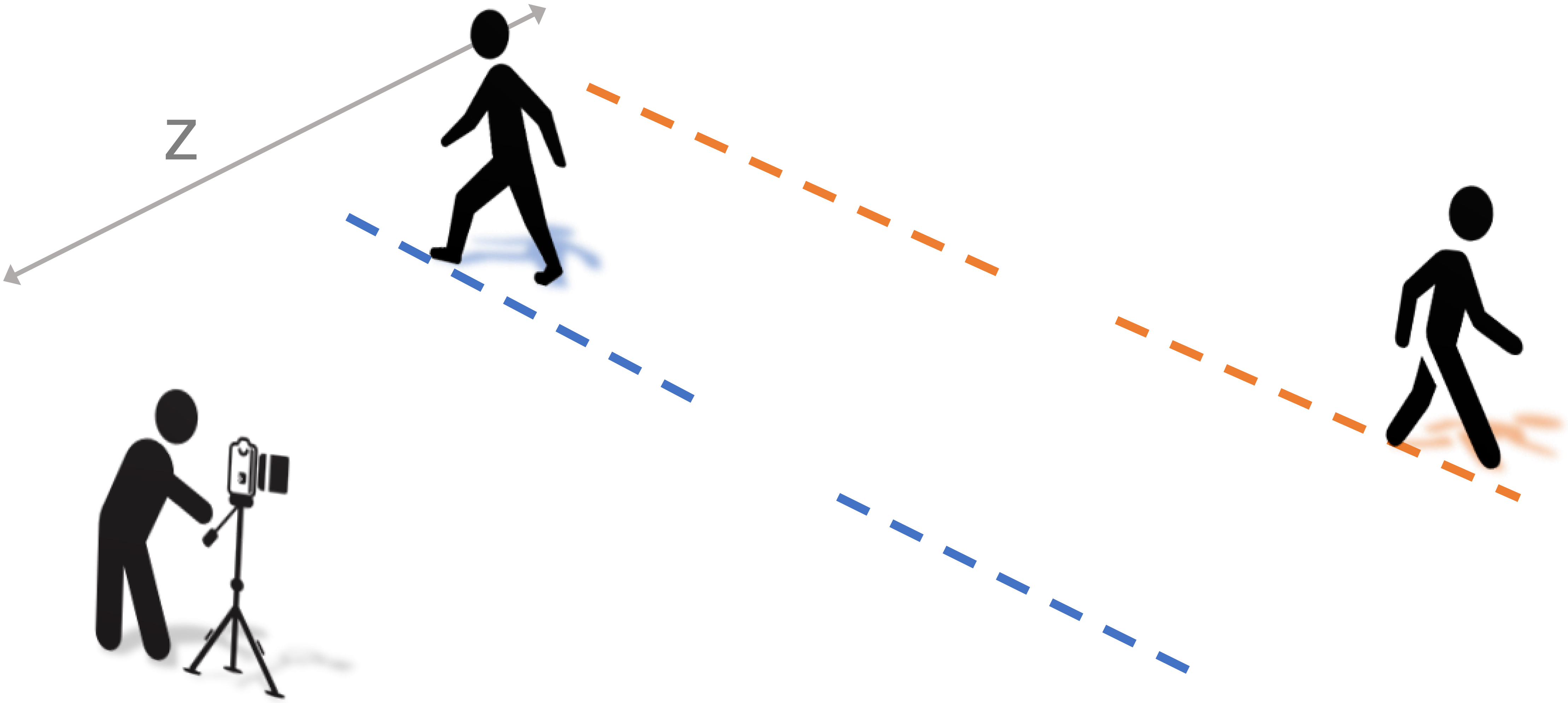
- In 2D, occlusion is hard to disambiguate

Insight #1

- In 2D, occlusion is hard to disambiguate

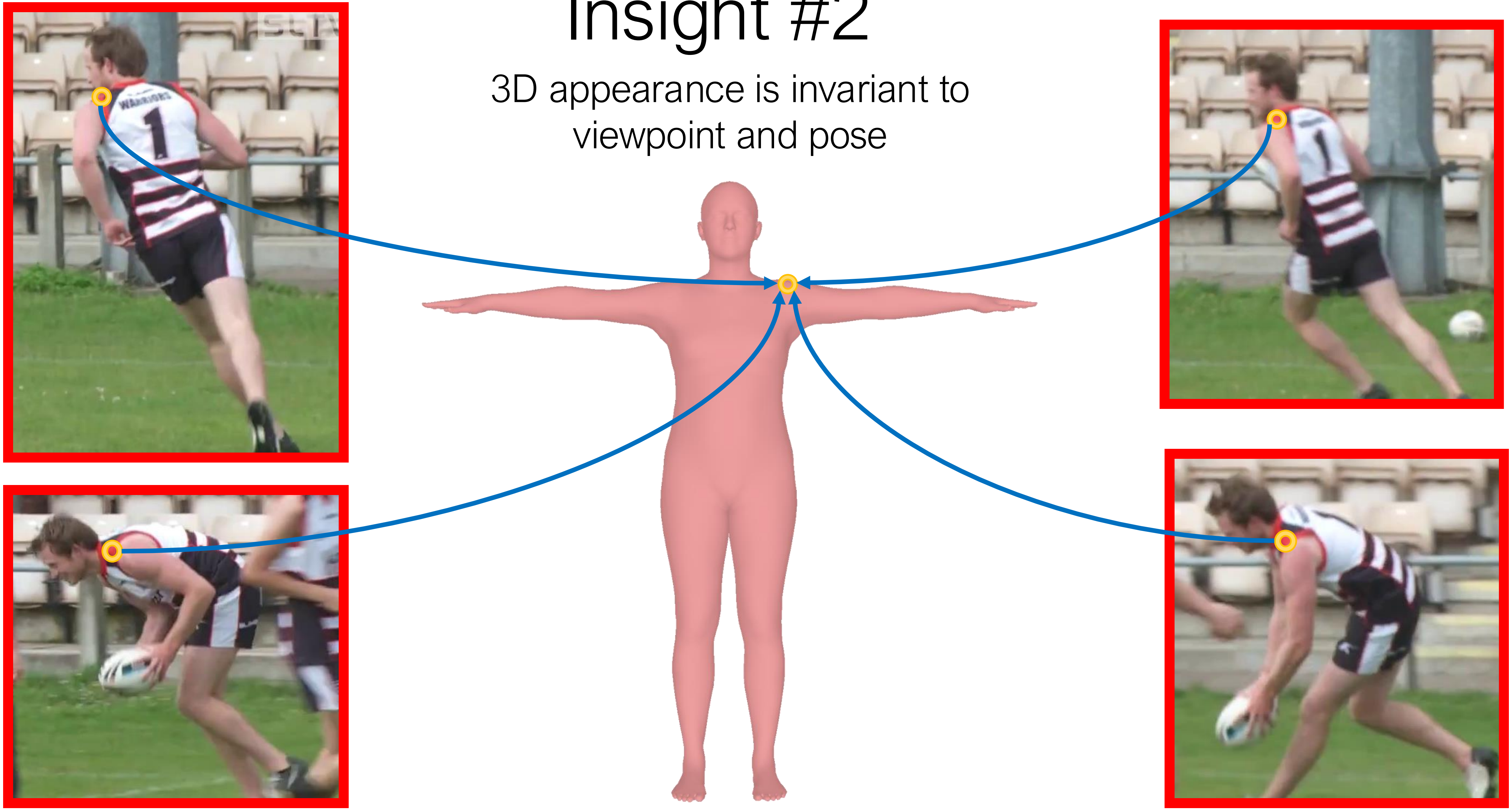
Insight #1

- In 2D, they overlap, but in 3D they don't!



Insight #2

3D appearance is invariant to viewpoint and pose




G.W.COX
MY JEWELLER
TED BAKER
LONDON
NIKE


Rio2016
Ω OMEGA
GO
0:55
KOR
SON
Yeon Jae
Rio2016

What are the next steps?

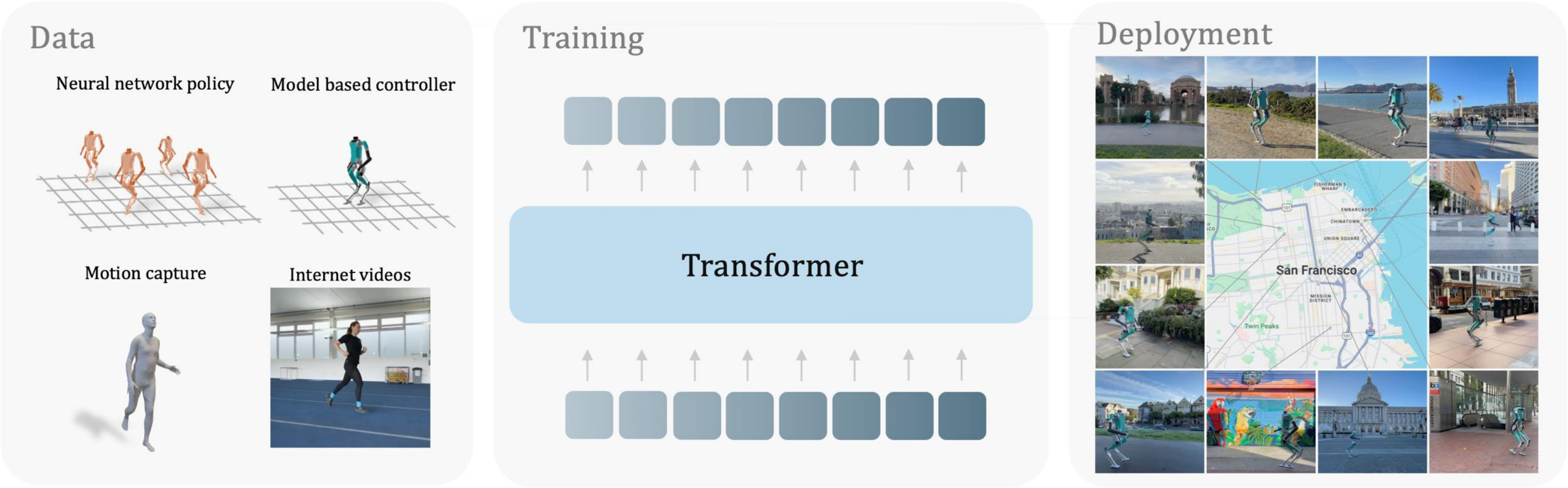
Humanoid Locomotion as Next Token Prediction

Ilija Radosavovic Bike Zhang Baifeng Shi Jathushan Rajasegaran Sarthak Kamat
Trevor Darrell Koushil Sreenath Jitendra Malik

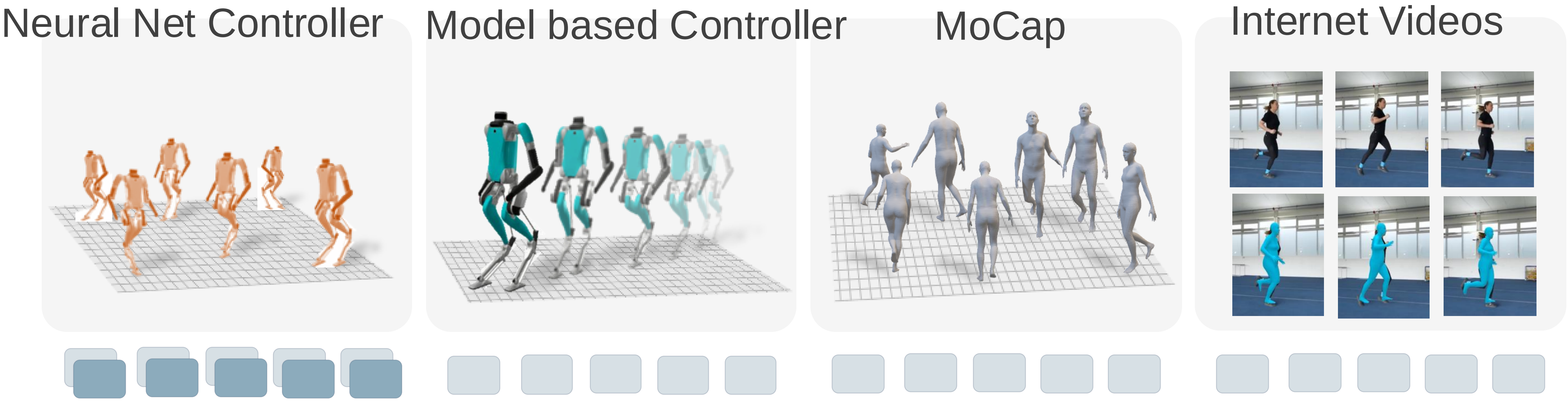
University of California, Berkeley


Berkeley
UNIVERSITY OF CALIFORNIA

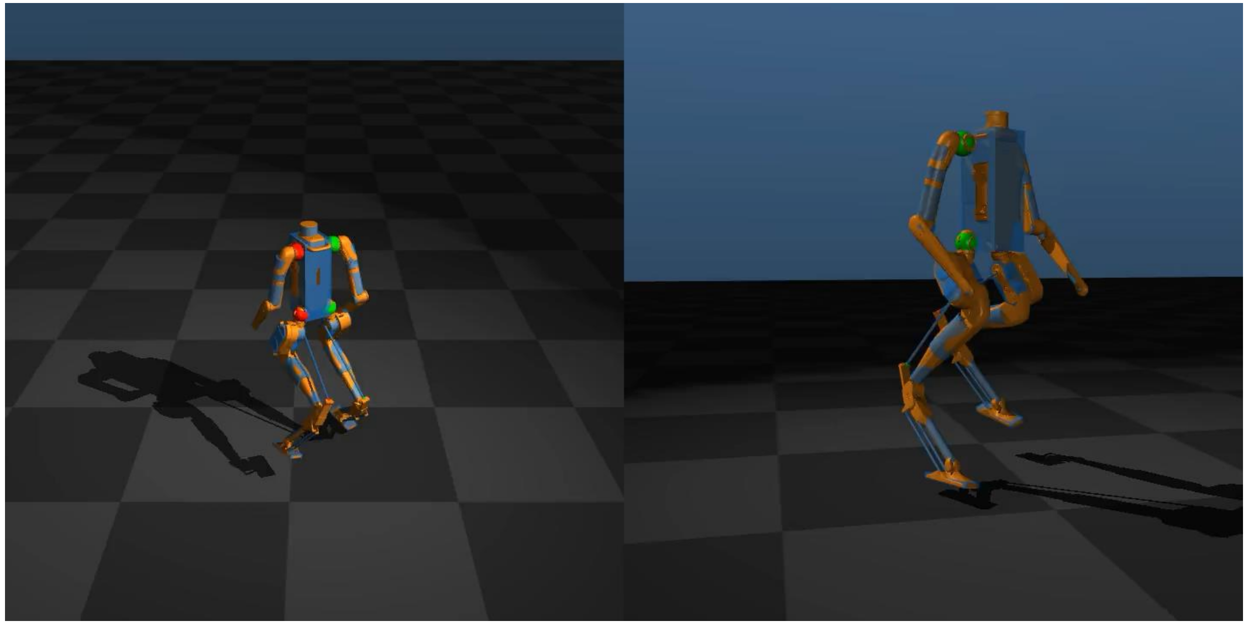
Humanoid Control as Next Token Prediction



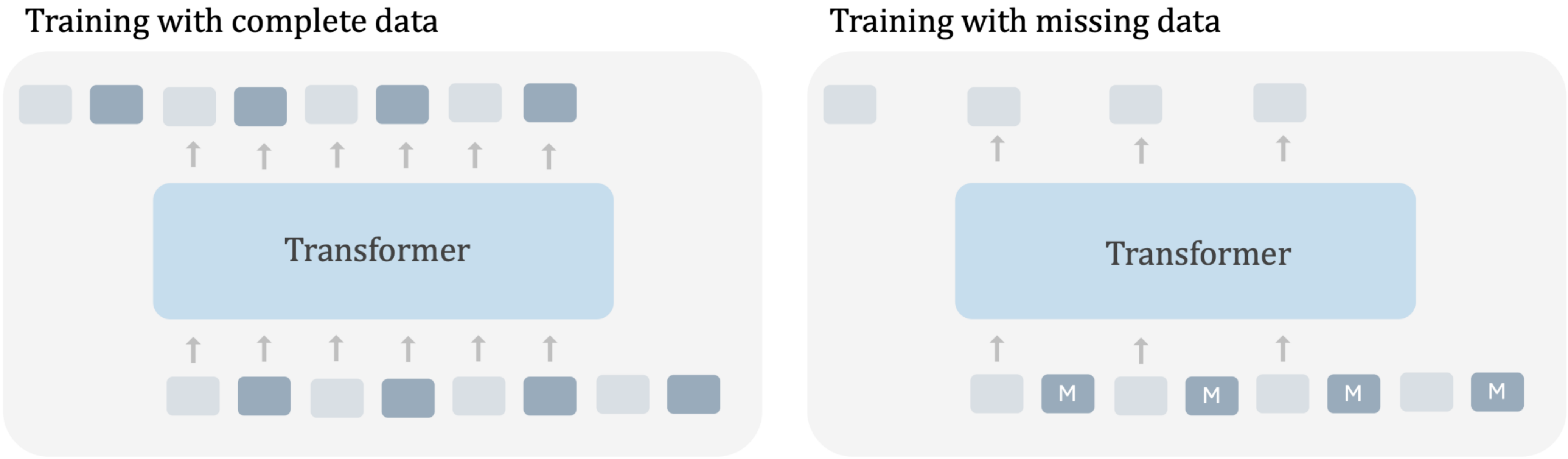
A Dataset of Walking Trajectories



Retarget YouTube Video in Simulation



Training with Missing Data



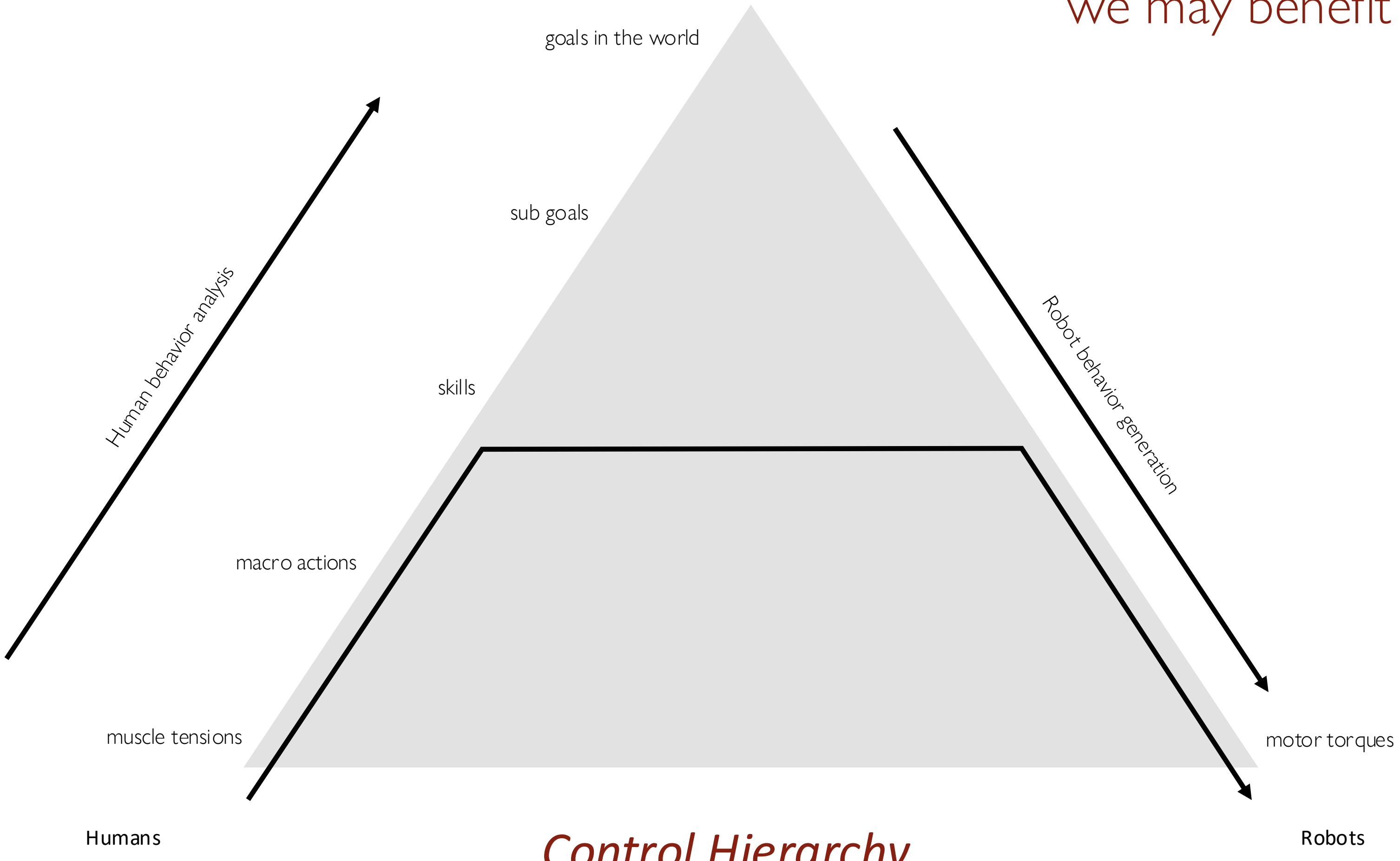
Learning at different abstraction levels

Credit: Saurabh Gupta, UIUC

Depending on the amount of gap between:

- goals,
- embodiment,
- what we can observe in videos

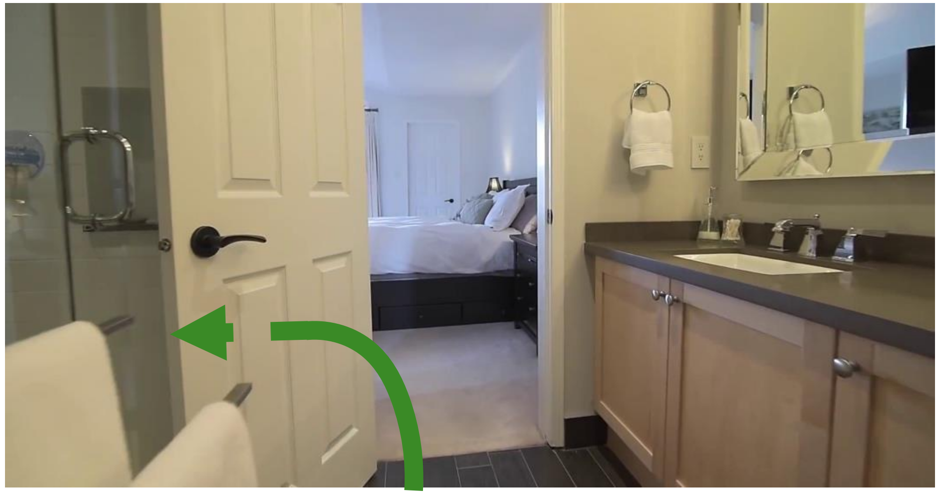
we may benefit from transfer at different levels.



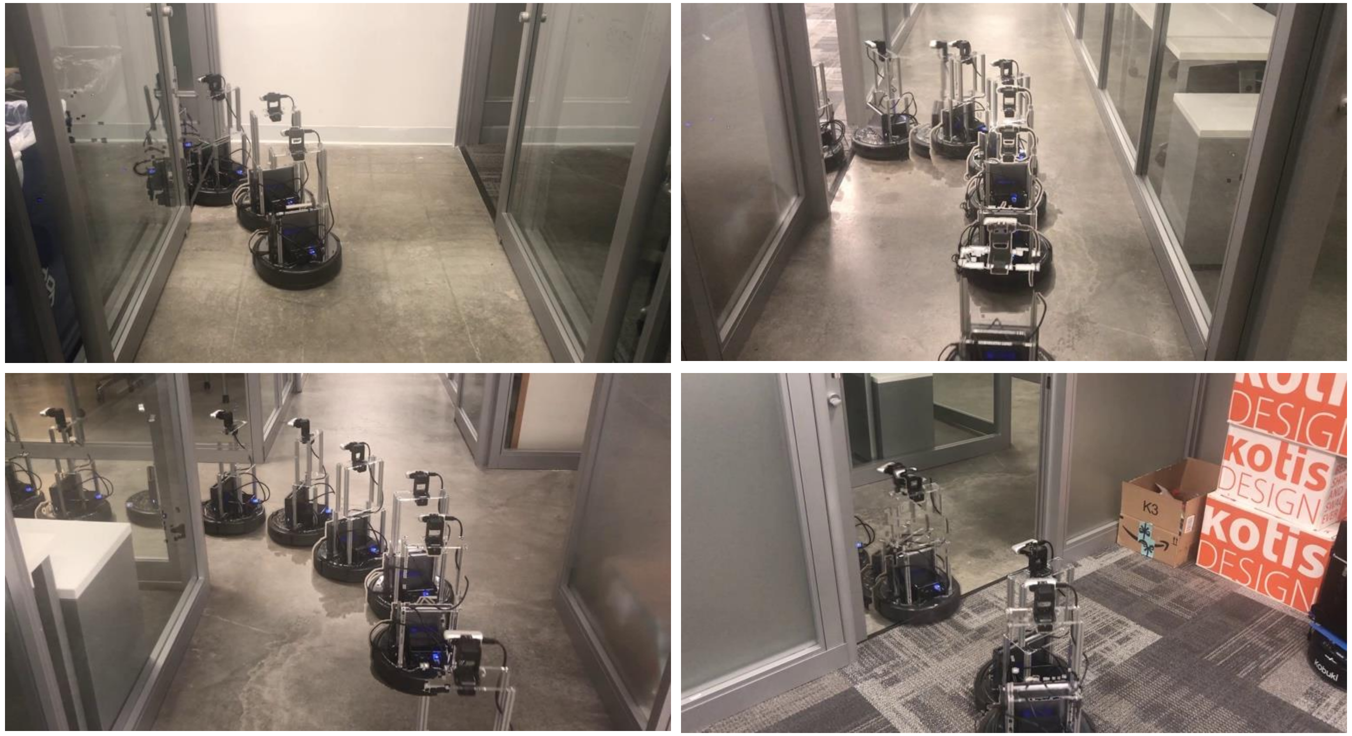
Control Hierarchy

We have found success with applying this idea to navigation

- Affordances (What can you do here)
- Subroutines (How)

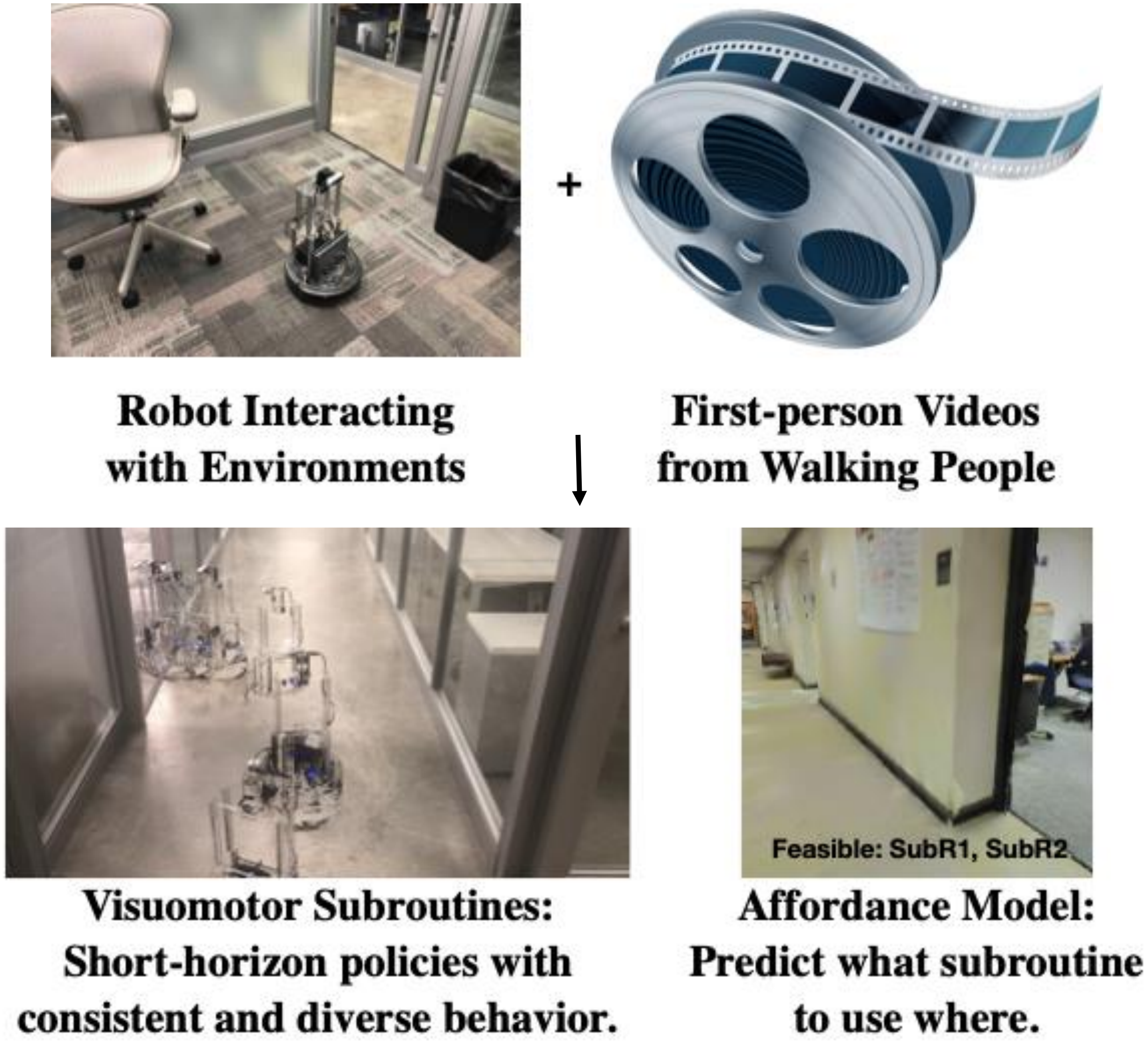


Turn left

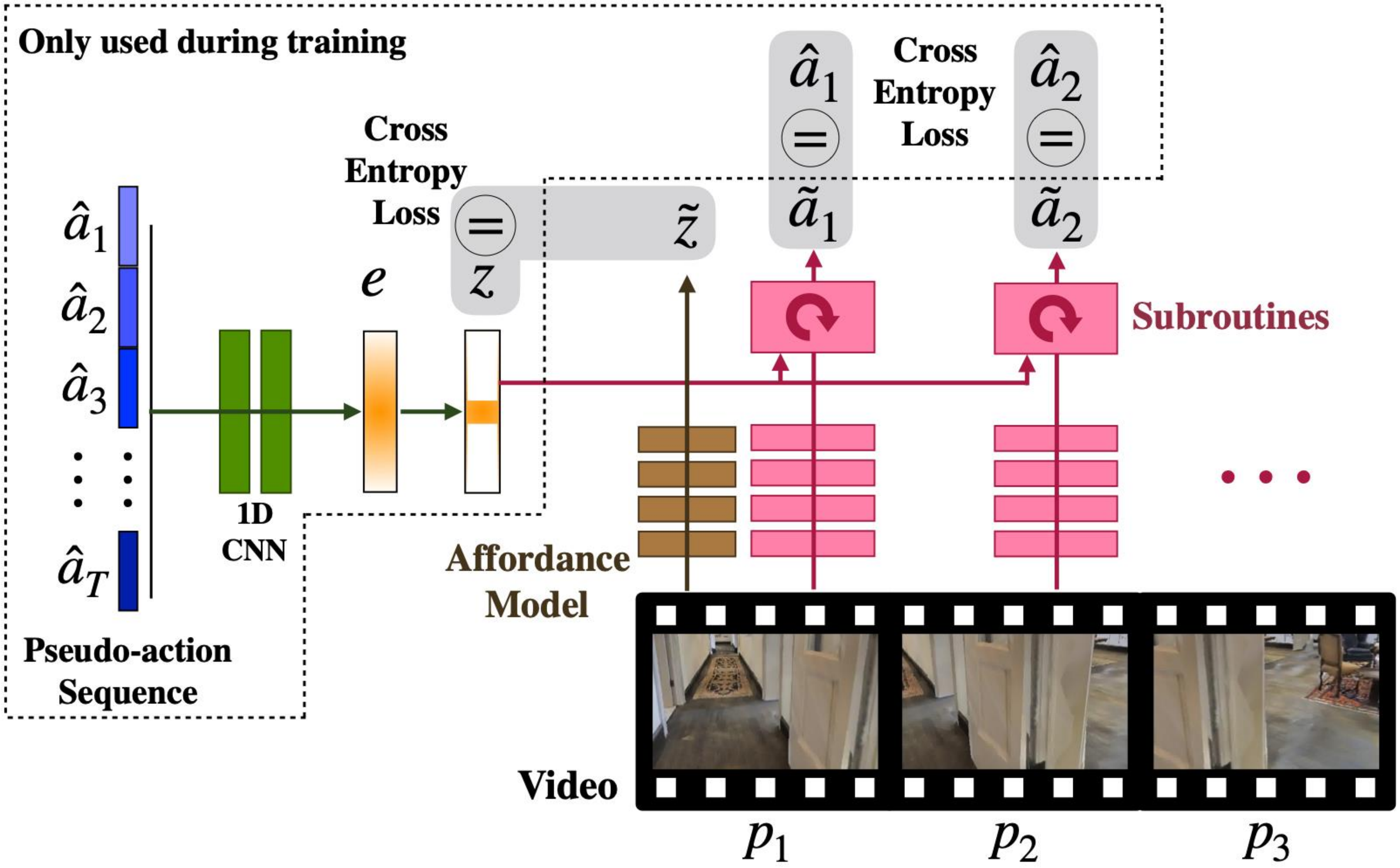


Kumar et. al. Learning Navigation Subroutines from Egocentric Videos CoRL

Learning Navigation by Watching Videos



A. Kumar, S. Gupta, J. Malik. *CoRL 2019.* Learning Navigation Subroutines by Watching Videos.



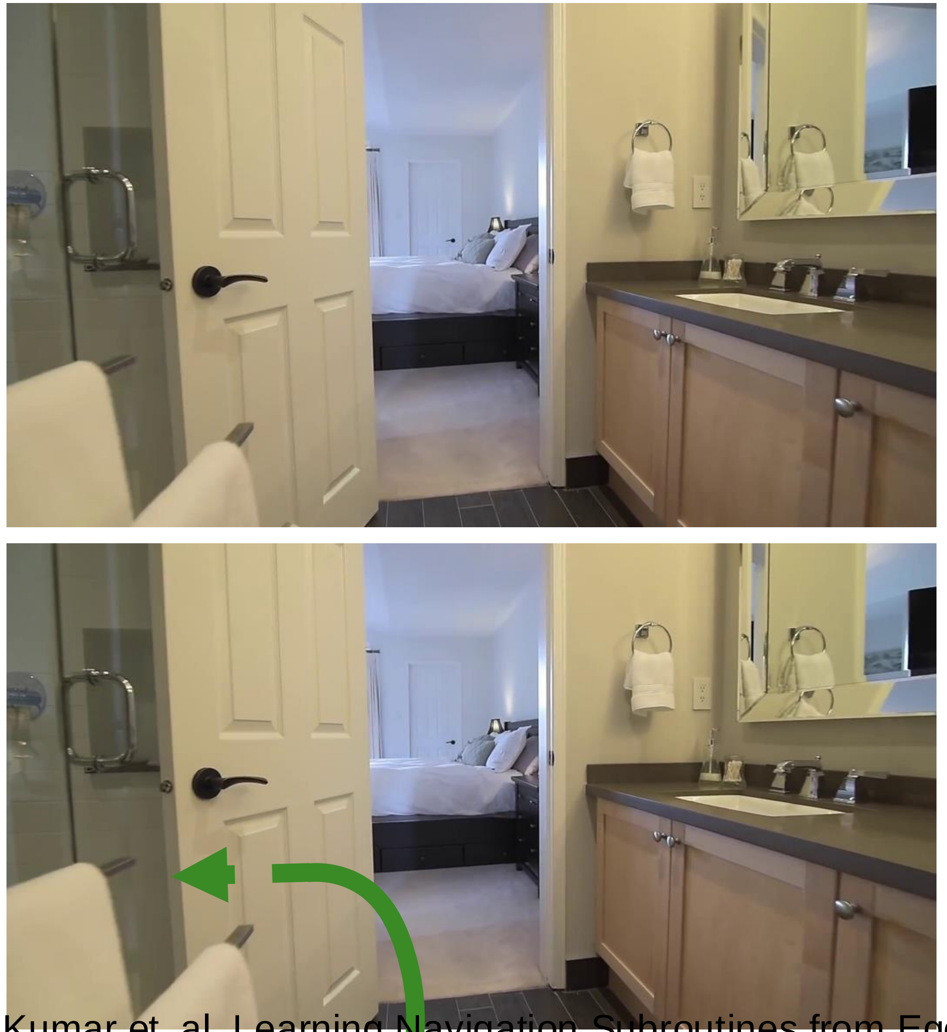
Only used during training
Cross Entropy Loss
$\hat{a}_1$
$\hat{a}_2$
$\hat{a}_3$
$\hat{a}_T$
Pseudo-action Sequence
1D CNN
e
z
$\tilde{z}$
Cross Entropy Loss
$\tilde{a}_1$
$\tilde{a}_2$
Subroutines
Affordance Model
Video
p_1
p_2
p_3

SubR1 - Turns Right

SubR3 - Turns Left

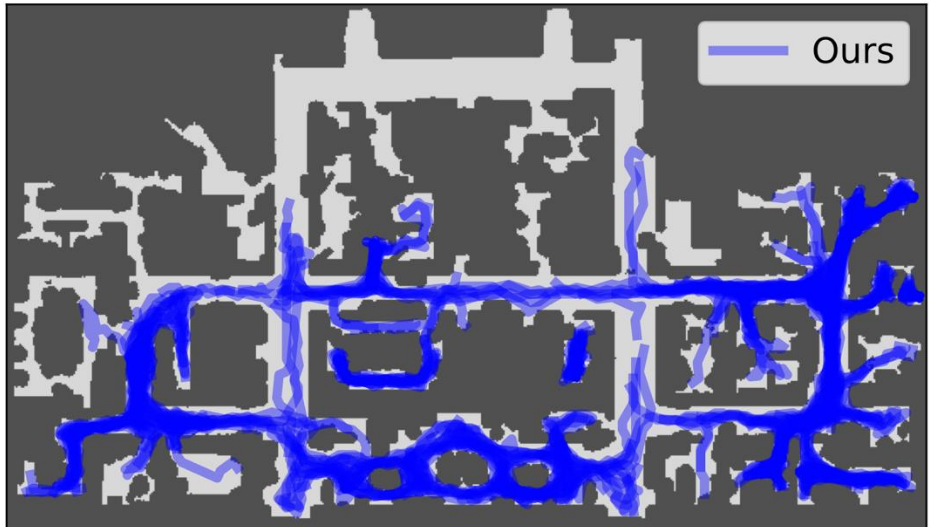
We have found success with applying this idea to navigation

- Affordances (What can you do here)
- Subroutines (How)

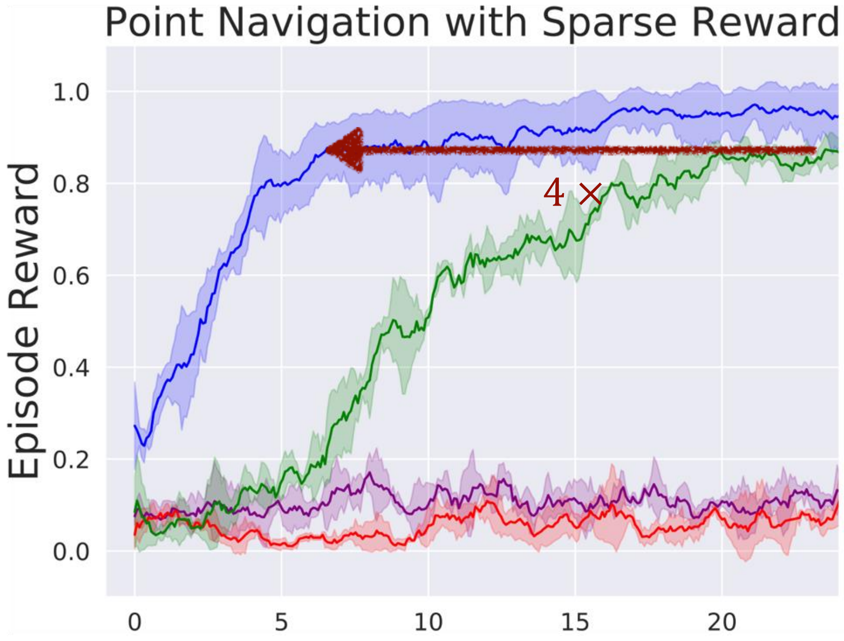


Kumar et. al. Learning Navigation Subroutines from Egocentric Videos. CoRL

a) Use subroutines and affordances for exploration



b) Use subroutines and affordances for for hierarchical RL



INSTRUCTIONS ON HOW TO CLIMB A STAIRCASE

No one will have failed to observe that frequently the floor bends in such a way that one part rises at a right angle to the plane formed by the floor and then the following section arranges itself parallel to the flatness, so as to provide a step to a new perpendicular, a process which is repeated in a spiral or in a broken line to highly variable elevations. Ducking down and placing the left hand on one of the vertical parts and the right hand upon the corresponding horizontal, one is in momentary possession of a step or stair. Each one of these steps, formed as we have seen by two elements, is situated somewhat higher and further than the one prior, a principle which gives the idea of a staircase, while whatever other combination, producing perhaps more beautiful or picturesque shapes, would be incapable of translating one from the ground floor to the first floor.

You tackle a stairway face on, for if you try it backwards or sideways, it ends up being particularly uncomfortable. The natural stance consists of holding oneself upright, arms hanging easily at the sides, head erect but not so much so that the eyes no longer see the steps immediately above, while one tramps up, breathing lightly and with regularity. To climb a staircase one begins by lifting that part of the body located below and to the right, usually encased in leather or deerskin, and which, with a few exceptions, fits exactly on the stair. Said part set down on the first step (to abbreviate we shall call it "the foot"), one draws up the equivalent part on the left side (also called "foot" but not to be confused with "the foot" cited above), and lifting this other part to the level of "the foot," makes it continue along until it is set in place on the second step, at which point the foot will rest, and "the foot" will rest on the first. (The first steps are always the most difficult, until you acquire the necessary coordination. The coincidence of names between the foot and "the foot" makes the explanation more difficult. Be especially careful not to raise, at the same time, the foot and "the foot.")

Having arrived by this method at the second step, it's easy enough to repeat the movements alternately, until one reaches the top of the staircase. One gets off it easily, with a light tap of the heel to fix it in place, to make sure it will not move until one is ready to come down.

Julio Cortazar